Artificial Intelligence

Transforming network operations with AI: How Swisscom built a network assistant using Amazon Bedrock

by Pablo García Benedicto, Jordi Montoliu Nerin, Rajesh Sripathi, and Ruben Merz | on 03 JUL 2025 | in Amazon Machine Learning, Artificial Intelligence, Customer Solutions, Generative AI | Permalink | Comments | Share

In the telecommunications industry, managing complex network infrastructures requires processing vast amounts of data from multiple sources. Network engineers often spend considerable time manually gathering and analyzing this data, taking away valuable hours that could be spent on strategic initiatives. This challenge led Swisscom, Switzerland's leading telecommunications provider, to explore how AI can transform their network operations.

Swisscom's Network Assistant, built on Amazon Bedrock, represents a significant step forward in automating network operations. This solution combines generative AI capabilities with a sophisticated data processing pipeline to help engineers quickly access and analyze network data. Swisscom used AWS services to create a scalable solution that reduces manual effort and provides accurate and timely network insights.

In this post, we explore how Swisscom developed their Network Assistant. We discuss the initial challenges and how they implemented a solution that delivers measurable benefits. We examine the technical architecture, discuss key learnings, and look at future enhancements that can further transform network operations. We highlight best practices for handling sensitive data for Swisscom to comply with the strict regulations governing the telecommunications industry. This post provides telecommunications providers or other organizations managing complex infrastructure with valuable insights into how you can use AWS services to modernize operations through AI-powered automation.

The opportunity: Improve network operations

Network engineers at Swisscom faced the daily challenge to manage complex network operations and maintain optimal performance and compliance. These skilled professionals were tasked to monitor and analyze vast amounts of data from multiple and decoupled sources. The process was repetitive and demanded considerable time and attention to detail. In certain scenarios, fulfilling the assigned tasks consumed more than 10% of their availability. The manual nature of their work presented several critical pain points. The data consolidation process from multiple network entities into a coherent overview was particularly challenging, because engineers had to navigate through various tools and systems to retrieve telemetry information about data sources and network parameters from extensive documentation, verify KPIs through complex calculations, and identify potential issues of diverse nature. This fragmented approach consumed valuable time and introduced the risk of human error in data interpretation and analysis. The situation called for a solution to address three primary concerns:

- Efficiency in data retrieval and analysis
- Accuracy in calculations and reporting
- Scalability to accommodate growing data sources and use cases

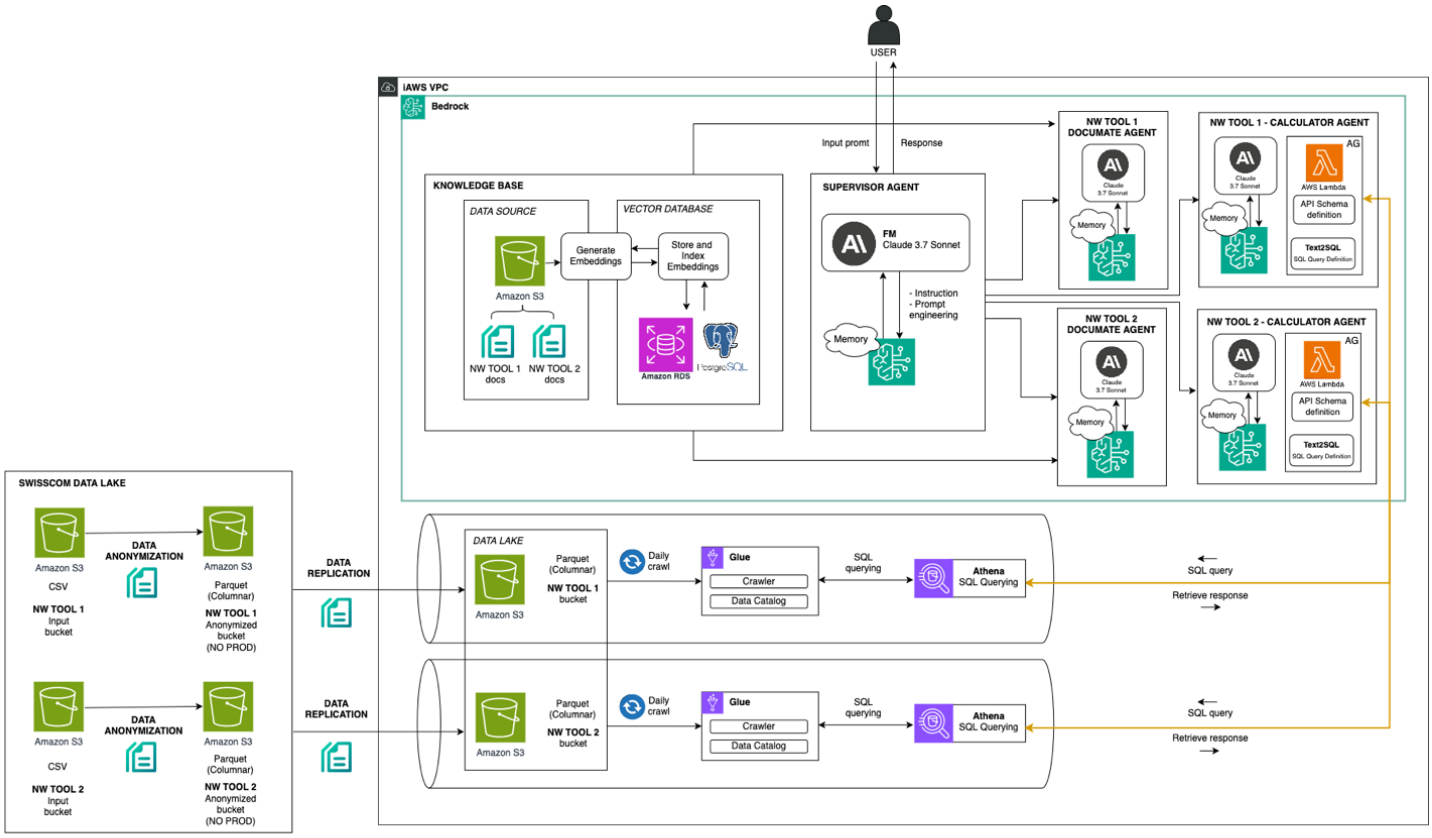
The team required a streamlined approach to access and analyze network data, maintain compliance with defined metrics and thresholds, and deliver fast and accurate responses to events while maintaining the highest standards of data security and sovereignty.

Solution overview

Swisscom's approach to develop the Network Assistant was methodical and iterative. The team chose Amazon Bedrock as the foundation for their generative AI application and implemented a Retrieval Augmented Generation (RAG) architecture using Amazon Bedrock Knowledge Bases to enable precise and contextual responses to engineer queries. The RAG approach is implemented in three distinct phases:

- **Retrieval** – User queries are matched with relevant knowledge base content through embedding models
- **Augmentation** – The context is enriched with retrieved information
- **Generation** – The large language model (LLM) produces informed responses

The following diagram illustrates the solution architecture.



The solution architecture evolved through several iterations. The initial implementation established basic RAG functionality by feeding the Amazon Bedrock knowledge base with tabular data and documentation. However, the Network Assistant struggled to manage large input files containing thousands of rows with numerical values across multiple parameter columns. This complexity highlighted the need for a more selective approach that could identify only the rows relevant for specific KPI calculations. At that point, the retrieval process wasn't

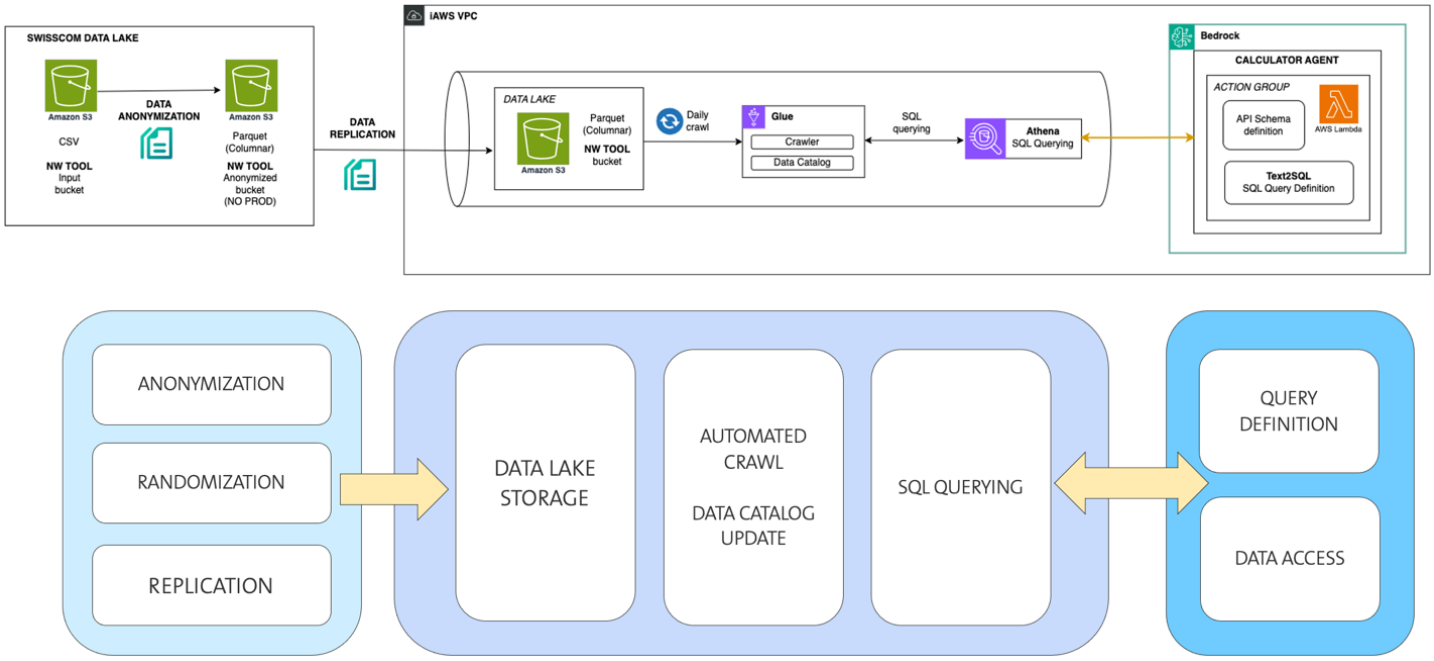
returning the precise number of vector embeddings required to calculate the formulas, prompting the team to refine the solution for greater accuracy.

Next iterations enhanced the assistant with agent-based processing and action groups. The team implemented AWS Lambda functions using Pandas or Spark for data processing, facilitating accurate numerical calculations retrieval using natural language from the user input prompt.

A significant advancement was introduced with the implementation of a multi-agent approach, using Amazon Bedrock Agents, where specialized agents handle different aspects of the system:

- **Supervisor agent** – Orchestrates interactions between documentation management and calculator agents to provide comprehensive and accurate responses.
- **Documentation management agent** – Helps the network engineers access information in large volumes of data efficiently and extract insights about data sources, network parameters, configuration, or tooling.
- **Calculator agent** – Supports the network engineers to understand complex network parameters and perform precise data calculations out of telemetry data. This produces numerical insights that help perform network management tasks; optimize performance; maintain network reliability, uptime, and compliance; and assist in troubleshooting.

This following diagram illustrates the enhanced data extract, transform, and load (ETL) pipeline interaction with Amazon Bedrock.



To achieve the desired accuracy in KPI calculations, the data pipeline was refined to achieve consistent and precise performance, which leads to meaningful insights. The team implemented an ETL pipeline with Amazon Simple Storage Service (Amazon S3) as the data lake to store input files following a daily batch ingestion approach, AWS Glue for automated data crawling and cataloging, and Amazon Athena for SQL querying. At this point, it became possible for the calculator agent to forego the Pandas or Spark data processing implementation.

Instead, by using Amazon Bedrock Agents, the agent translates natural language user prompts into SQL queries. In a subsequent step, the agent runs the relevant SQL queries selected dynamically through analysis of various input parameters, providing the calculator agent an accurate result. This serverless architecture supports scalability, cost-effectiveness, and maintains high accuracy in KPI calculations. The system integrates with Swisscom's on-premises data lake through daily batch data ingestion, with careful consideration of data security and sovereignty requirements.

To enhance data security and appropriate ethics in the Network Assistant responses, a series of guardrails were defined in Amazon Bedrock. The application implements a comprehensive set of data security guardrails to protect against malicious inputs and safeguard sensitive information. These include content filters that block harmful categories such as hate, insults, violence, and prompt-based threats like SQL injection. Specific denied topics and sensitive identifiers (for example, IMSI, IMEI, MAC address, or GPS coordinates) are filtered through manual word filters and pattern-based detection, including regular expressions (regex). Sensitive data such as personally identifiable information (PII), AWS access keys, and serial numbers are blocked or masked. The system also uses contextual grounding and relevance checks to verify model responses are factually accurate and appropriate. In the event of restricted input or output, standardized messaging notifies the user that the request can't be processed. These guardrails help prevent data leaks, reduce the risk of DDoS-driven cost spikes, and maintain the integrity of the application's outputs.

Results and benefits

The implementation of the Network Assistant is set to deliver substantial and measurable benefits to Swisscom's network operations. The most significant impact is time savings. Network engineers are estimated to experience 10% reduction in time spent on routine data retrieval and analysis tasks. This efficiency gain translates to nearly 200 hours per engineer saved annually, and represents a significant improvement in operational efficiency. The financial impact is equally impressive. The solution is projected to provide substantial cost savings per engineer annually, with minimal operational costs at less than 1% of the total value generated. The return on investment increases as additional teams and use cases are incorporated into the system, demonstrating strong scalability potential.

Beyond the quantifiable benefits, the Network Assistant is expected to transform how engineers interact with network data. The enhanced data pipeline supports accuracy in KPI calculations, critical for network health tracking, and the multi-agent approach provides orchestrated and comprehensive responses to complex queries out of user natural language.

As a result, engineers can have instant access to a wide range of network parameters, data source information, and troubleshooting guidance from an individual personalized endpoint with which they can quickly interact and obtain insights through natural language. This enables them to focus on strategic tasks rather than routine data gathering and analysis, leading to a significant work reduction that aligns with Swisscom SRE principles.

Lessons learned

Throughout the development and implementation of the Swisscom Network Assistant, several learnings emerged

that shaped the solution. The team needed to address data sovereignty and security requirements for the solution, particularly when processing data on AWS. This led to careful consideration of data classification and compliance with applicable regulatory requirements in the telecommunications sector, to make sure that sensitive data is handled appropriately. In this regard, the application underwent a strict threat model evaluation, verifying the robustness of its interfaces against vulnerabilities and acting proactively towards securitization. The threat model was applied to assess doomsday scenarios, and data flow diagrams were created to depict major data flows inside and beyond the application boundaries. The AWS architecture was specified in detail, and trust boundaries were set to indicate which portions of the application trusted each other. Threats were identified following the STRIDE methodology (Spoofing, Tampering, Repudiation, Information disclosure, Denial of service, Elevation of privilege), and countermeasures, including Amazon Bedrock Guardrails, were defined to avoid or mitigate threats in advance.

A critical technical insight was that complex calculations involving significant data volume management required a different approach than mere AI model interpretation. The team implemented an enhanced data processing pipeline that combines the contextual understanding of AI models with direct database queries for numerical calculations. This hybrid approach facilitates both accuracy in calculations and richness in contextual responses.

The choice of a serverless architecture proved to be particularly beneficial: it minimized the need to manage compute resources and provides automatic scaling capabilities. The pay-per-use model of AWS services helped keep operational costs low and maintain high performance. Additionally, the team's decision to implement a multi-agent approach provided the flexibility needed to handle diverse types of queries and use cases effectively.

Next steps

Swisscom has ambitious plans to enhance the Network Assistant's capabilities further. A key upcoming feature is the implementation of a network health tracker agent to provide proactive monitoring of network KPIs. This agent will automatically generate reports to categorize issues based on criticality, enable faster response time, and improve the quality of issue resolution to potential network issues. The team is also exploring the integration of Amazon Simple Notification Service (Amazon SNS) to enable proactive alerting for critical network status changes. This can include direct integration with operational tools that alert on-call engineers, to further streamline the incident response process. The enhanced notification system will help engineers address potential issues before they critically impact network performance and obtain a detailed action plan including the affected network entities, the severity of the event, and what went wrong precisely.

The roadmap also includes expanding the system's data sources and use cases. Integration with additional internal network systems will provide more comprehensive network insights. The team is also working on developing more sophisticated troubleshooting features, using the growing knowledge base and agentic capabilities to provide increasingly detailed guidance to engineers.

Additionally, Swisscom is adopting infrastructure as code (IaC) principles by implementing the solution using AWS CloudFormation. This approach introduces automated and consistent deployments while providing version control of infrastructure components, facilitating simpler scaling and management of the Network Assistant solution as it grows.

Conclusion

The Network Assistant represents a significant advancement in how Swisscom can manage its network operations. By using AWS services and implementing a sophisticated AI-powered solution, they have successfully addressed the challenges of manual data retrieval and analysis. As a result, they have boosted both accuracy and efficiency so network engineers can respond quickly and decisively to network events. The solution's success is aided not only by the quantifiable benefits in time and cost savings but also by its potential for future expansion. The serverless architecture and multi-agent approach provide a solid foundation for adding new capabilities and scaling across different teams and use cases.As organizations worldwide grapple with similar challenges in network operations, Swisscom's implementation serves as a valuable blueprint for using cloud services and AI to transform traditional operations. The combination of Amazon Bedrock with careful attention to data security and accuracy demonstrates how modern AI solutions can help solve real-world engineering challenges.

As managing network operations complexity continues to grow, the lessons from Swisscom's journey can be applied to many engineering disciplines. We encourage you to consider how Amazon Bedrock and similar AI solutions might help your organization overcome its own comprehension and process improvement barriers. To learn more about implementing generative AI in your workflows, explore Amazon Bedrock Resources or contact AWS.

Additional resources

For more information about Amazon Bedrock Agents and its use cases, refer to the following resources:

- Generative AI for telecom
- Best practices for building robust generative AI applications with Amazon Bedrock Agents – Part 1
- Best practices for building robust generative AI applications with Amazon Bedrock Agents – Part 2

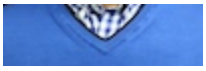
About the authors



Pablo García Benedicto is an experienced Data & AI Cloud Engineer with strong expertise in cloud hyperscalers and data engineering. With a background in telecommunications, he currently works at Swisscom, where he leads and contributes to projects involving Generative AI applications and agents using Amazon Bedrock. Aiming for AI and data specialization, his latest projects focus on building intelligent assistants and autonomous agents that streamline business information retrieval, leveraging cloud-native architectures and scalable data pipelines to reduce toil and drive operational efficiency.



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